Digitally signed by AHMAD QASIM MOHAMMAD Hasan
Date: 05/18/2026 00:16:09 +00:00

TID — Flush+Reload Side-Channel Attack Test | Live Report

Ahmad Qasim Mohammad Hassan | ORCID: 0009-0001-4360-0802

Generated : 2026-05-16 21:39:37 UTC
CPU : model name : AMD EPYC 9B14
Kernel : 6.14.11
Compiler : gcc (GCC) 14.3.0

TOOLS USED IN THIS ATTACK TEST

RDTSC : Real CPU cycle counter — measures attacker reload latency
CLFLUSH : Attacker's tool — evicts target from cache (Step 1)
CLFLUSHOPT : TID v2 defense — evicts cache after wipe (Step 3 defense)
REP STOSQ : Compiler-resistant wipe — cannot be optimized away
LFENCE : Prevents speculative execution around measurements
MFENCE : Full memory barrier — ensures ordering
Samples : 1000 per scenario — median used to remove outliers

HOW THE ATTACK WORKS — 4 STEPS

```
Step 1 [Flush]  : Attacker calls CLFLUSH on target address
                  -> evicts target from L1/L2/L3 cache
Step 2 [Use]    : Victim uses sensitive data
                  -> data loads back into cache
Step 3 [Wipe]   : TID is called
                  -> v1: wipes RAM only (cache stays loaded!)
                  -> v2: wipes RAM + evicts cache with CLFLUSHOPT
Step 4 [Reload] : Attacker measures reload time via RDTSC
                  -> ~78  cycles = Cache HIT  = data was there = LEAK
                  -> ~312 cycles = Cache MISS = cache empty    = SAFE
```

KEY FUNCTIONS IN SOURCE CODE

```
  clflush_attacker()       -> Step 1: attacker evicts cache
  tid_v1_wipe()            -> Step 3a: REP STOSQ only (no CLFLUSHOPT)
  tid_v2_wipe()            -> Step 3b: REP STOSQ + CLFLUSHOPT + MFENCE
  attacker_measure_reload()-> Step 4: RDTSC measurement with LFENCE
```

```
SCENARIO 1 — TID v1 (REP STOSQ only, NO CLFLUSHOPT)
  Attack tool: CLFLUSH + RDTSC | 1000 samples | median reported
```

```
  Samples         : 1000
  Median latency  : 78 cycles  (measured by RDTSC)
  Mean latency    : 71 cycles
  Interpretation  : ~78 cycles = L1 Cache HIT time => data still in cache
  Verdict         : Cache HIT ⚠  —   40141b:   0f ae 3b                 clflush (%rbx)
  40141e:      0f ae f0                 mfence
  40142d:      0f ae f0                 mfence
  401438:      0f ae e8                 lfence
  40143b:      0f 31                    rdtsc
  401441:      0f ae e8                 lfence
  401447:      0f ae e8                 lfence
  40144a:      0f 31                    rdtsc
  40144c:      0f ae e8                 lfence
  4015bb:      0f ae 3b                 clflush (%rbx)
  4015be:      0f ae f0                 mfence
  4015cd:      0f ae f0                 mfence
  4015d8:      0f ae e8                 lfence
  4015db:      0f 31                    rdtsc
  4015e1:      0f ae e8                 lfence
ATTACK SUCCEEDED (data leaked)
```

```
SCENARIO 2 — TID v2 (REP STOSQ + CLFLUSHOPT + MFENCE + LFENCE)
  Attack tool: CLFLUSH + RDTSC | 1000 samples | median reported
```

```
  Samples         : 1000
  Median latency  : 312 cycles  (measured by RDTSC)
  Mean latency    : 332 cycles
```

Interpretation : ~312 cycles = RAM access time => cache is empty
Verdict : Cache MISS ✓ — ATTACK DEFEATED (cache empty)

══

FINAL COMPARISON

══

v1 Median (Cache HIT — data leaked): 78 cycles
v2 Median (Cache MISS — safe): 312 cycles
Ratio (v2/v1): 4.0x
Conclusion: TID v2 SUCCESSFULLY defeats Flush+Reload ✓

══

DISASSEMBLY PROOF — instructions confirmed in -O2 binary

══